THE PALAEOMAGNETISM OF THE KAINOZOIC BASALTS OF VICTORIA

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(Communicated by S. K. Runcorn)

(Received 1957 May 21)

Summary

The average geomagnetic pole position consistent with the magnetization of the Newer Volcanics of Victoria, which are for the most part of Pleistocene age, coincides with the present *geographic* pole just as do all pole determinations from Miocene and later rock formations from the Northern Hemisphere. During this time it appears that the Earth's magnetic field has approximated, on average, to that of a geocentric axial dipole and that the land masses concerned have maintained their present positions. In contrast, the pole determined from Lower Tertiary Volcanics of Victoria does not agree with the geographic pole nor with equivalent determinations from the Northern Hemisphere, suggesting that the relative positions of the continents may have been different in the past.

1. *Introduction.*—Some crude observations of the directions of natural remanent magnetization of Kainozoic basalts from several parts of Australia have been reported previously, and it is interesting that in some cases these are in the opposite sense to the present geomagnetic field. This was noticed by Mercanton (**1**) in three specimens from Queensland, by Rodgers (**2**) in four specimens from New England (New South Wales), and by Almond, Clegg and Jaeger (**3**) in one bore-core specimen from Tasmania. Local anomalies in the geomagnetic field shown by field variometer surveys carried out by Rayner (**4, 5**) over basalts in Central New South Wales and New England indicate that in some of these the direction of permanent magnetization is also reversely polarized. Although these basalts in Queensland, New South Wales and Tasmania are known to be Kainozoic, their position within this period is in dispute.

In Victoria the basalts form part of the standard Kainozoic sequence of Australia and it is possible to separate the flows into two groups whose geological ages for these purposes are satisfactorily defined. Because of this, our observations have been concentrated in these two groups, and they represent the first systematic survey of palaeomagnetic directions in rock formations of Kainozoic age in the Southern Hemisphere.

In the Northern Hemisphere, where a great deal of attention has been paid to the palaeomagnetism of rock formations of this age, specimens from Miocene and younger formations show reversed or normal polarizations with roughly equal frequency, and the directions of magnetization, regardless of sign, oscillate around the direction of the present geocentric axial dipole field; in contrast, the directions of magnetization observed in Lower Tertiary formations in the Northern Hemisphere, although exhibiting both polarities, do not concur with the axial dipole field. The Kainozoic basalts of Victoria show precisely the same characteristics.

It is convenient to interpret palaeomagnetic results in terms of pole positions. Assuming that the Earth's magnetic field when averaged out over several thousands of years approximates to a geocentric axial dipole, the direction of the central

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dipole in whose field the rocks became magnetized may be calculated; this assumption may be referred to as the dipole hypothesis. The poles referred to here are the points where the axis of this central dipole intersect the Earth's surface.

The pole positions (Fig. 11) consistent with the direction of the Newer Volcanics (mostly of Pleistocene age) are coincident within the errors with the present geographic poles, and it appears that so far as this region is concerned the Earth, during the past 10^6 years, has on average behaved as if it were uniformly magnetized with its axis of magnetization coincident with the axis of spin. Equivalent determinations from the Northern Hemisphere also concur with the geographic poles, and it appears that the geomagnetic field has held this uniform character for the past 20×10^6 years. It is unlikely that important land-movements between the Northern continents and Australia have occurred during the past 10^6 years.

The poles consistent with the magnetization of the Older Volcanics (Lower Tertiary) do not agree with the geographic poles nor with equivalent determinations from the Northern Hemisphere (Fig. 12). This disagreement could have arisen through relative movements between Australia and these continents prior to the Pleistocene.

2. *Geology.*—The Kainozoic volcanics are divided into two main groups, the Newer and Older Volcanics (Singleton (**6**)). Both groups are olivine basalts with acid differentiates and are distinguished from one another by important petrological differences (Edwards (**7, 8**)). Edwards has recognized two iron minerals, ilmenite and magnetite.

The Newer Volcanics have a more or less continuous outcrop to the west of the meridian of Melbourne. They overlie the marine Tertiaries of the region and cannot be older than Pliocene. In fact, it is certain that the bulk of the lavas belong to the Pleistocene, but it is possible that some of the earlier flows could be Pliocene. The youngest flows are geologically recent. The Newer Volcanics have in places been faulted, but tilting of the lava field has been negligible.

The Older Volcanics, sometimes called the Narracan Volcanics, outcrop mainly to the east of Melbourne and are interbedded with non-marine sediments which form part of the Tertiary sequence in Victoria. In the area sampled they are believed to represent a single stratigraphic entity which is certainly Lower Tertiary in age and probably Eocene. This Lower Tertiary sequence has been deformed by Late Tertiary movements, but these have not been extensive and steeply dipping beds are restricted to the neighbourhood of faults and monoclinal fields. There is no metamorphism. The present discontinuous outcrops have resulted from erosion subsequent to this deformation.

3. *Experimental method*

3.1. *Measurement.*—Hand specimens have been orientated by standard geological techniques. Cylinders have been machined from these with non-magnetic tools, orientation being preserved during the process. The directions of magnetization and initial susceptibility have been measured on the magnetometer described previously (**9**). Inclusive of orientation and measurement errors, each direction determination is, at $P = 0{\cdot}95$, within 10° of the true direction, that is, the probable error is 5° (**10**). Individual determinations of intensity and susceptibility are accurate to 5 per cent.

3.2. *Sampling procedure.*—The basalts have suffered surface weathering in recent times. Since this alters the magnetic properties, only fresh material from

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sea-cliffs and quarries at least 20 feet below the surface has been obtained. Thirty rock slides have been examined to test for incipient weathering of the iron-minerals which may not be detectable in hand specimens. In all cases the outlines of the iron-minerals are clean and sharp, and there is no indication of alteration.

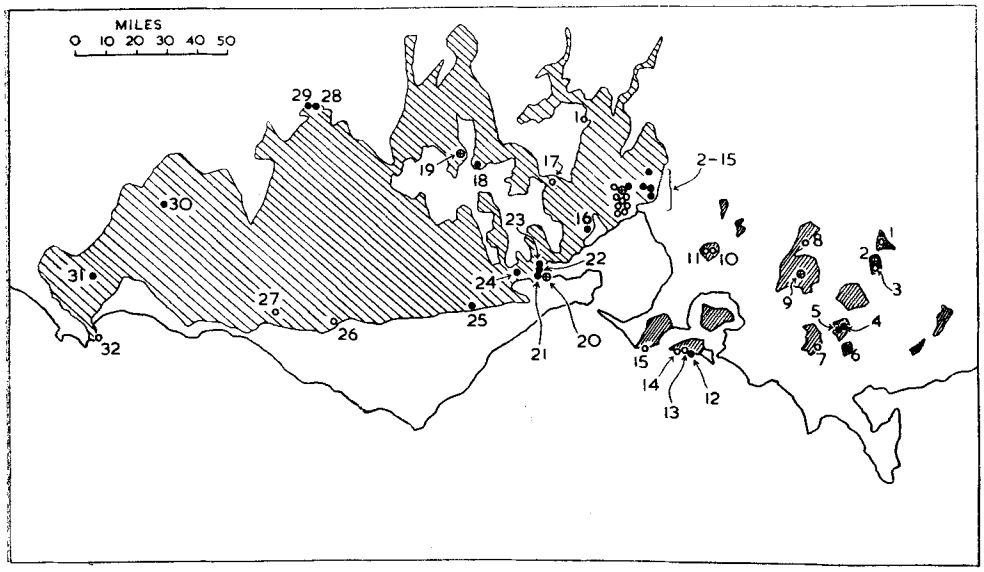


FIG. 1.—*Sampling sites in the Victorian basalts.*

The Newer and Older Volcanics are distinguished by broad and narrow shading respectively. Full dots indicate sites with reversed magnetization, normal sites are shown by open circles, and sites with mixed polarity by circles with a cross. The numbering corresponds to the numbering in Figs. 2, 3 and 6.

The method used here of conducting a palaeomagnetic survey of volcanic formations with optimum sample economy has been described previously (**10**). For the general survey, samples have been taken from as many sites as possible spread over the outcrop, with two or three samples of each to minimize experimental error. In this way inhomogeneities due to the secular change of the geomagnetic field and geological tilting are averaged out, and a wider regional picture is obtained than would otherwise be the case if the same number of samples were distributed among fewer sites. Errors due to geological tilting have been reduced to a minimum by using only those sites at which the beds are known to be flat-lying or dipping by no more than 5°. The effect of this will, in any case, be lost when results from many localities are averaged, and the present horizontal will be a close approximation to the horizontal at the time of deposition.

In the Newer Volcanics two samples have been taken at each of thirty sites, and two sites have been sampled in detail. The coverage is approximately 10,000 square miles. In the Older Volcanics only fifteen suitable sites were available, and because of this three samples were taken at each. The area covered is about 5,000 square miles.

4. *The natural permanent magnetization*

4.1. *The directions of magnetization.*—Allowing for experimental error, there is always close agreement between samples from the same site. The degree of uniformity at two sites sampled in detail is illustrated in Fig. 2. At all except five localities the polarities also agree. Contiguous specimens with opposed

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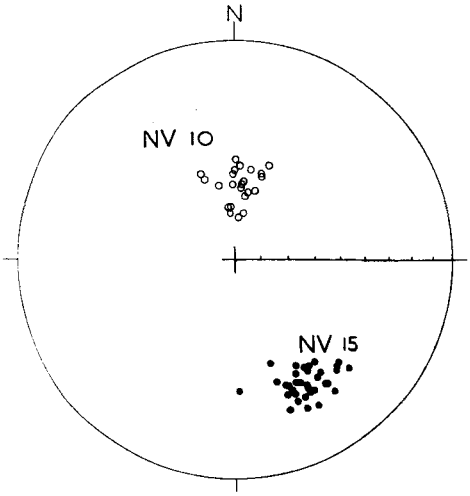


Fig. 2

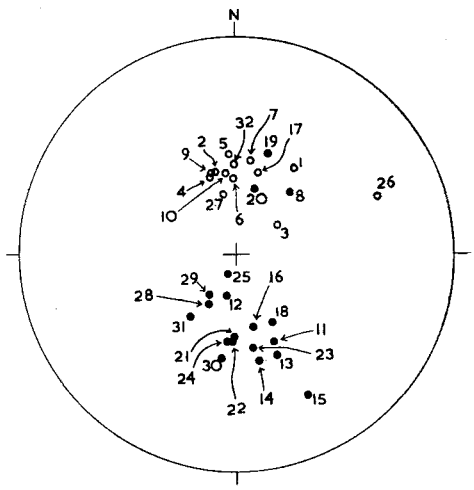


Fig. 3

FIG. 2.—*The within-site variation in directions of magnetization at two sites in the Newer Volcanics.*

Here and in Figs. 3, 4 and 5 an equal-area Schmidt net is used, the plane of the projection is the present horizontal, the full dots are north-seeking directions plotted on the lower hemisphere (reversed), and the open circles are north-seeking directions in the upper hemisphere (normal). Geographic North is marked. Footscray Quarry (NV 10), $N=23$, $D=0°{\cdot}3$, $I=-61°{\cdot}8$, $\kappa_1=90{\cdot}1$, $\alpha\,(P=0{\cdot}05)=3°{\cdot}2$. Darebin Quarry (NV 15), $N=34$, $D=149°{\cdot}6$, $I=+34°{\cdot}9$, $\kappa_1=88{\cdot}0$, $\alpha\,(P=0{\cdot}05)=2°{\cdot}7$. κ_1 and α are Fisher's precision parameter and circle of confidence (12).

FIG. 3.—*The mean directions at thirty-two sites in the Newer Volcanics.*

Conventions as in Fig. 2, except that sites with mixed polarity, 8, 20, 19, are plotted without respect to sign on the upper hemisphere.

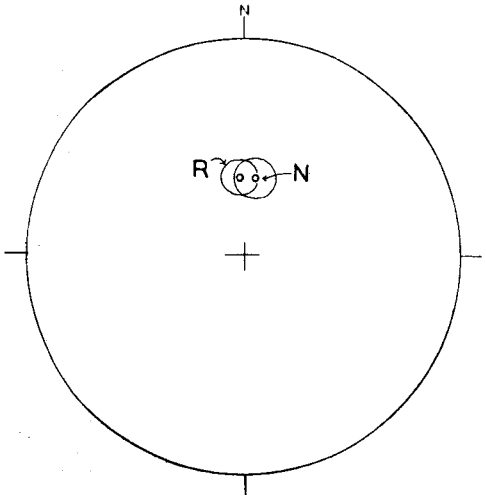


Fig. 4

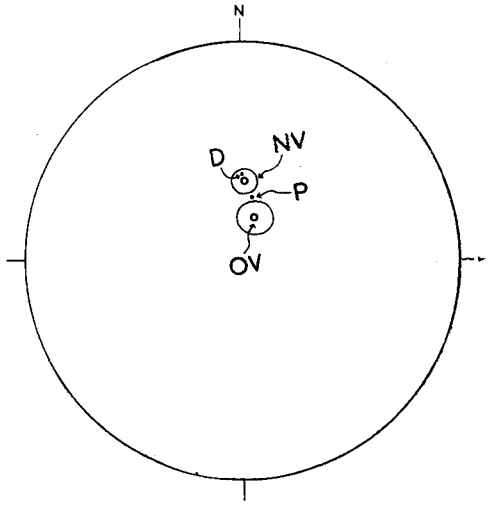


Fig. 5

FIG. 4.—*The mean direction of the Normal* (N) *and Reversed* (R) *sites in the Newer Volcanics.*

The circles are the error circles at $P=0{\cdot}05$. Both directions are in the upper hemisphere, north-seeking in the case of mean of the normal sites, and south-seeking for the reversed site mean direction.

FIG. 5.—*Mean directions of the Newer* (NV) *and Older* (OV) *Volcanics.*

These mean directions have been calculated regardless of sign. D is the dipole field, and P is the direction of the present field in Victoria. The circles of error are at $P=0{\cdot}05$.

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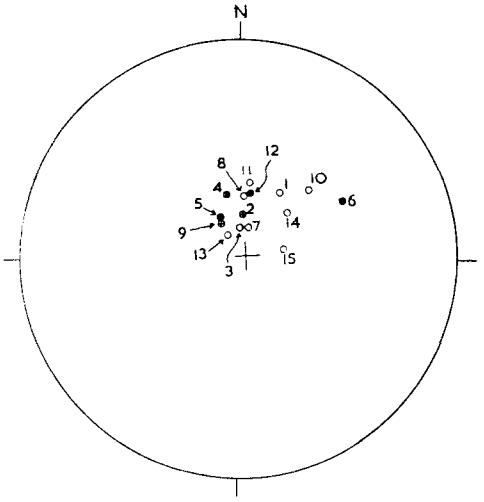


FIG. 6.—*Mean directions in fifteen sites in the Older Volcanics.*
The conventions are as in Fig. 3, except that all directions are now plotted on the upper hemisphere. 9 and 12 are mixed localities.

polarity have been noted previously in these basalts by Rayner (private communication) and in the basalts of Cape Kawajiri in Japan by Asami (**11**).

The mean directions at thirty-two sites in the Newer Volcanics are plotted in Fig. 3. Thirteen sites have the polarity of the present field (north-seeking polarization upwards) and sixteen are of opposite polarity, the two groups being exactly 180° apart (Fig. 4). Three sites have mixed polarity. It is impracticable to arrange the sites in chronological order and any association of reversed directions with particular periods of time cannot be detected. The overall mean direction, regardless of sign, is parallel to the geocentric axial dipole field (Table I, Fig. 5).

TABLE I

The directions of magnetization in the Victorian Basalts

Rock Groups		N	D	I	κ_2	α	β	ψ
Newer Volcanics	Reversed	16	177°·1	+60°·0	443·2	6·7	31	25·1
	Normal	13	8°·2	−59°·6	302·1	8·0	35	23·7
	Total	32	3°·4	−59°·8	833·3	4·8	37	23·0
Older Volcanics		15	17°·0	−72°·9	430·5	6·8	35	23·1

(N is the number of sampling sites, D and I are the declination and inclination, κ_2 is the overall precision (**10**), α is the Fisher error circle (**12**) at $P=0{\cdot}05$ calculated from a modified formula (**10**), β is the between-site dispersion, ψ is the semi-angle of the cone around the mean direction containing 95 per cent of the mean site directions.)

The mean site directions in the Older Volcanics are given in Fig. 6. Nine sites have negative inclination, four are positive, and two sites have mixed polarity. The data is insufficient to test the exactness of these reversals. The average

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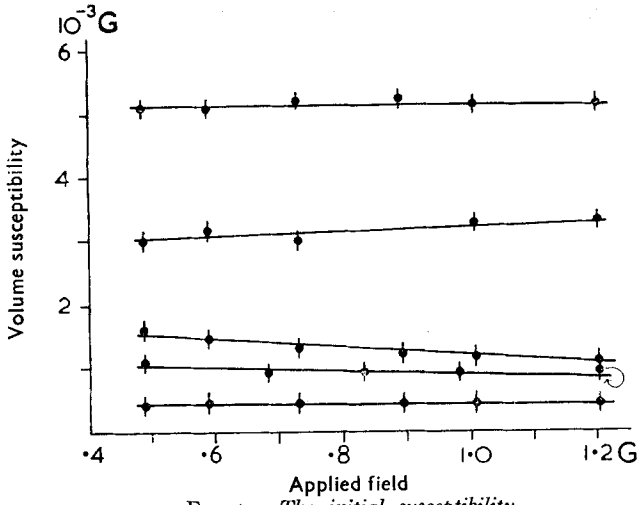


FIG. 7.—*The initial susceptibility.*

The susceptibility is defined as the ratio of the intensity of magnetization induced by an external field to the strength of that field. The values obtained do not vary appreciably in fields of between 0·49 and 1·21 oersted.

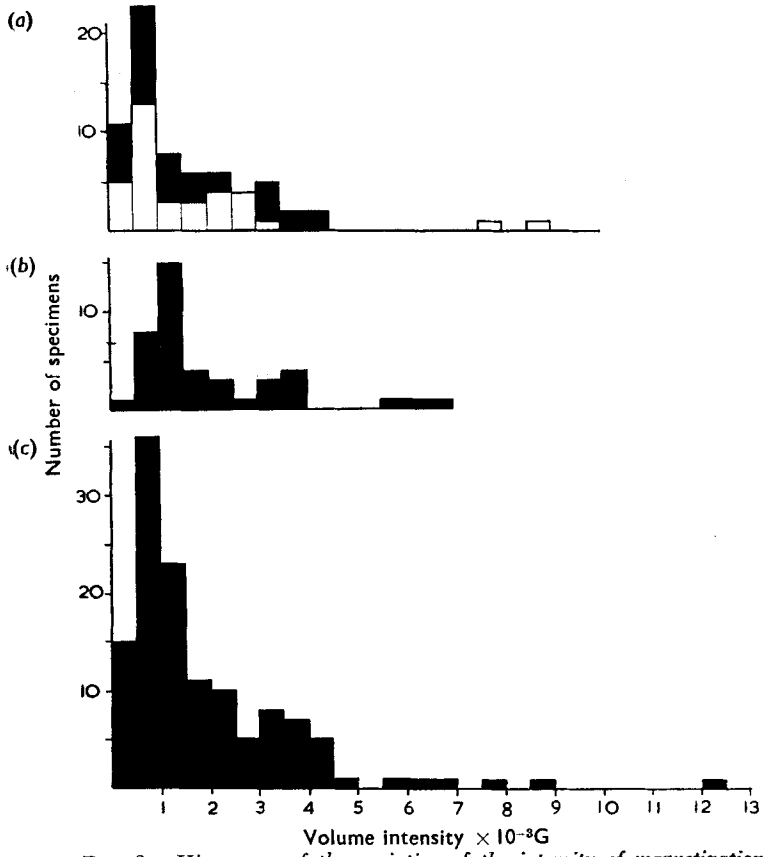


FIG. 8.—*Histograms of the variation of the intensity of magnetization.*

(*a*) the Newer Volcanics, normally magnetized specimens in black, reversed ones left blank; (*b*) the Older Volcanics; (*c*) all basalt specimens, including some from Queensland and New South Wales.

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direction of the whole group, regardless of sign, is given in Table I and Fig. 5; the inclination is steeper than in the Newer Volcanics and the declination is easterly.

In addition to the reversals in the Victoria basalts and those previously described from Queensland (**1**), Northern and Central New South Wales (**2, 4, 5**) and Tasmania (**3**), the authors have also observed reversals in the basalts of Toowoomba (Queensland) and Berrima (New South Wales). Clearly reversed magnetizations are a characteristic feature of the Kainozoic basalts of Australia.

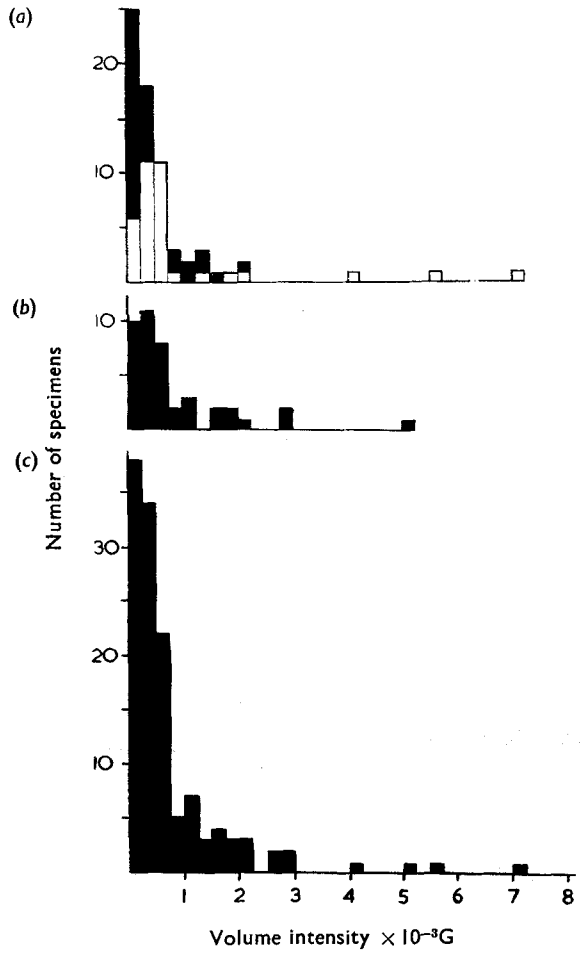


FIG. 9.—*Histograms of the variations of susceptibility.*

(*a*) the Newer Volcanics, normally magnetized specimens in black, reversed ones left blank; (*b*) the Older Volcanics; (*c*) all basalt specimens, including some from Queensland and New South Wales.

4.2. *The intensity of magnetization and initial susceptibility.*—In D.C. fields of up to 1·21 oersted the induced moment in basalt specimens is proportional to the applied field (Fig. 7). This initial susceptibility has been measured in all specimens and the values are indicative of the amount of magnetic material present.

The distributions of intensity of magnetization and susceptibility in specimens from a rock formation are of an irregular type, and at present qualitative comparison only are possible. The intensity and susceptibility of the Newer Volcanics on

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average appear to be lower than in the Older group and the ratio of these is comparable (Figs. 8, 9, 10). In the Newer Volcanics the susceptibility is somewhat greater in reversed than in normal specimens, whereas the intensities are similar, so that in normal samples the ratio I/K is somewhat higher.

4.3. *Stability.*—The following points provide evidence that the present directions of magnetization are the directions of the original thermo-remanent magnetization.

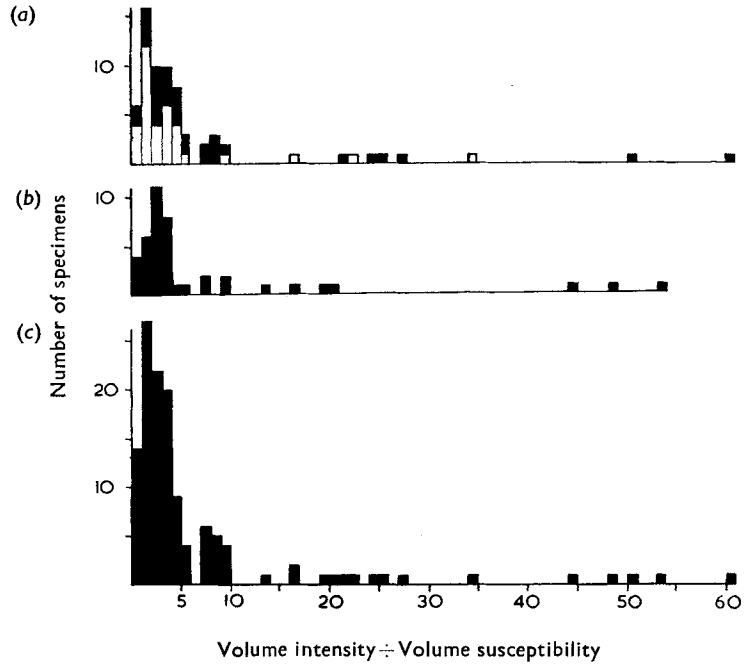


FIG. 10.—*Histograms of the variation of the ratio of the intensity of magnetization to the susceptibility.*

(*a*) the Newer Volcanics, normally magnetized specimens in black, reversed ones left blank; (*b*) the Older Volcanics; (*c*) all basalt specimens, including some from Queensland and New South Wales.

(*a*) Separate specimens from the same site and different sites in the same area give approximately parallel directions, and re-measurement after storage for a year at an angle to the Earth's field gives repeatable results.

(*b*) The mean direction of both groups which are plotted in Fig. 5 differs significantly from the present field, suggesting the directions have been stable over periods of the order of tens of years.

(*c*) The occurrence of reversals in specimens which are 10^5–10^6 years old in the case of the Newer Volcanics, and more than 20×10^6 years old in the Older Volcanics, indicates stability over these periods of time.

(*d*) As more data on the ratio of intensity to susceptibility accumulates, it is becoming increasingly apparent that where the initial directions of magnetization have been shown by various stability tests to have remained unchanged over long periods of geological time, I/K has the value of 1 or more. In rocks which have lost their initial direction of magnetization, this ratio rarely exceeds a few tenths and is usually very much less, which is presumably a result of the progressive replacement of the initial magnetization by a much weaker secondary isothermal

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magnetization. This is illustrated in Table II. The values in the Victorian basalts fall usually between 1·5 and 10 and are similar to those in other rocks with stable directions of magnetization, indicating that a high proportion of the initial magnetization has been retained.

TABLE II

The ratio of intensity to susceptibility in magnetically stable and unstable rocks

	Rock types	No. of determinations	I/K
Stable	Tertiary basalt from Iceland (**13**)	13	3·0
	Pre-Cambrian banded very fine grained sandstone from the Torridonian Sandstone Series of Scotland (**14**)	9	0·8–3·0
	Carboniferous red varvoid sediments from the Kuttung Series of New South Wales*	5	1–4
	Carboniferous toscanite lava from the Kuttung Series of New South Wales*	5	1–5
	Mesozoic dolerite from Tasmania (**9**)	60	2–10
Unstable	Permian basalt from New South Wales*	17	0·2–0·05
	Carboniferous horneblende andesite from the Kuttung Series of New South Wales*	6	0·1–0·4
	Pre-Cambrian green siltstones from the Torridonian Sandstone Series*	4	0·1–0·01

* Irving (unpublished work).

5. *Geophysical interpretation.*—Four items of geophysical interest arise out of these data.

5.1. *The dipole field direction in the Newer Volcanics.*—In the Newer Volcanics the average direction of magnetization is parallel to the present geocentric dipole field, and the poles of the axis of the mean geocentric dipole in whose field the rocks became magnetized coincide with the present geographic poles (Fig. 11, Table III). Three other pole determinations from Miocene and younger formations from the Northern Hemisphere are also given in Fig. 11. Many more

TABLE III

The pole positions determined from the directions of magnetization of the Victorian basalts

	Sampling Area		Pole Location		δm	δp
	Latitude	Longitude	Latitude	Longitude		
Newer Volcanics	38°·0S	143°·5E	86°·3S	102°·1E	7°·2	5°·5
Older Volcanics	38°·0S	145°·5E	66°·8S	122°·7E	12°·1	10°·8

δp is the error in the direction of colatitude and δm is the error in the perpendicular direction, both at $P=0{\cdot}05$.

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equivalent determinations have been given by Hospers (**15**), and all coincide within the errors with the geographic poles. It appears that since the Middle Tertiary the geomagnetic field, when viewed from four widely scattered places in the Earth's surface, has, on average, been similar to that of a geocentric axial dipole.

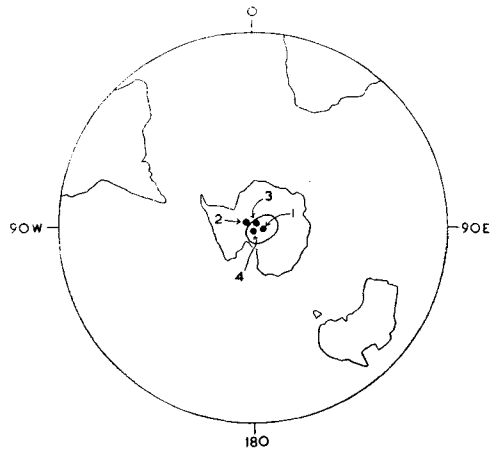


FIG. 11.—*The position of the mean geomagnetic pole during and since Miocene times.*

The poles are numbered as follows: (1) the Newer Volcanics of Victoria; (2) the recent lavas of Mount Etna (**17, 15**); (3) the reversed specimens from the Miocene to Pliocene Columbia River lavas (**18**); (4) the Miocene lavas of Iceland (**15**). In this and Fig. 12 the projection is an equal-area Schmidt net.

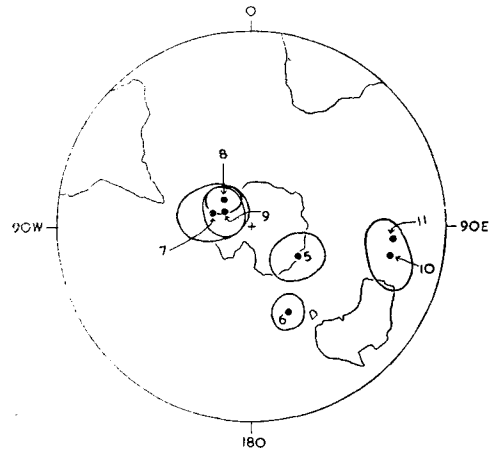


FIG. 12.—*The position of the mean geomagnetic pole in Lower Tertiary and Cretaceous times.*

The poles are numbered as follows: (5) the Older Volcanics of Victoria; (6) the Tasmanian dolerites (**9**) (Jurassic or Cretaceous); (7) the Oligocene of France (**19**); (8) the Eocene of Northern Ireland (**20**); (9) the Cretaceous of the United States (**21**); (10) the Deccan Traps, Cretaceous to Eocene, of India (**22**); (11) the Deccan Traps (**23**).

5.2. *Reversals of magnetization.*—Reversals of magnetization are a characteristic feature of the Kainozoic basalts of Australia and are comparable to those found in lavas of a similar age from the Northern Hemisphere. Three possible causes for these are considered.

(*a*) Special reversals of the Neel type (**16**).

(*b*) Reversals in polarity of the main geomagnetic field.

(*c*) Reversals resulting from variable stability.

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Suppose that by means of mechanism (*a*) or (*b*), all the magnetizations are initially reversed. In the former case the field has, since the early Tertiary, had its present sense; in the latter it has been reversed until quite recently. Normal directions could then arise in those cases where the magnetization is unstable by the imposition of a secondary isothermal magnetization by the present field which masks the reversed direction. If this were true, the ratio I/K in normal specimens would be much less than for reversed specimens. In fact, the opposite is the case (Fig. 10).

Two points lend support to the case for special self-reversals (*a*): the occurrence of either polarity at the same site, and the higher I/K values for normal specimens. Both these features may, however, be qualitatively explained without appeal to Neel-type mechanisms.

Inversion of specimens by a mistake in orientation is possible despite the care taken to avoid this in the orientation processes. At a site it has always been the intention to sample from a single flow only, but several flows, although not recognized, may be represented. A specimen from the upper part of a flow may have suffered re-heating by over-lying flows whereas one from the central parts may be unaffected. The possibility that specimens from the same site cooled down at different times between which the field reversed, although probably an uncommon occurrence, cannot be dismissed.

Most rocks possess a weak isothermal magnetization directed along the present dipole field, although in these basalts it must be small compared to the initial thermo-remanence. This reinforces a normal magnetization but opposes a reversed one. Another possibility is that the field intensity when the direction is normal may be greater than when reversed. Either of these may account for the higher I/K values in normal rocks.

In the Western District of Victoria the samples are widely scattered, and normal and reversed sites are apparently randomly scattered (Fig. 1). However, around Melbourne and Geelong, where sampling has been more intense, areas of common polarity can be recognized—for instance in S.E. Melbourne eight quarries have normal magnetization, whereas five quarries in N.W. Melbourne and four quarries in the Geelong area are consistently reversed. This uniformity of distribution in areas of dense sampling would, if anything, support the hypothesis of field reversals, but it is clearly impossible from the present data to decide between alternatives (*a*) and (*b*). In any case both processes may have been operative.

5.3. *The direction of magnetization of the Older Basalts.*—The average inclination of the Older Volcanics is greater than that in the Newer Volcanics and the declination is easterly (Table I). The pole positions are in the Davis Straits between Greenland and Baffin Island, and in Banzare Land in the Australian sector of Antarctica (Table III, Fig. 12). These poles are intermediate in position between those for the Newer Volcanics and the Tasmanian dolerites between which they are intermediate in time. Equivalent determinations from elsewhere in the world are added to Fig. 12. The poles from the European and North American data covering a period from Cretaceous to Oligocene fall between 13° and 23° of the geographic pole in longitudes between 47° and 97°, and do not differ significantly from one another. The equivalent poles from India and Australia do not agree with one another, are much further from the geographic pole and are in approximately opposite longitudes; the co-latitudes range between 62° and 23° and longitudes between 102° and 157°E. Three possible explanations for this are put forward.

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(*a*) That inadequate geological dating invalidates these comparisons, and that these pole discrepancies are due to the occurrence of geomagnetic polar wandering relative to the Earth as a whole between the time of deposition of the rock formations here compared. The effect of such polar wandering, if it occurred, should be present in the European and North American results; in fact, it is not (Fig. 12) (see note added in proof).

(*b*) That Australia and India have undergone movements with respect to each other and to Europe and North America during Cretaceous and Lower Tertiary times. This conclusion is contingent on the dipole hypothesis and cannot be avoided if this is true. Translation along a line of latitude producing a change in longitude does not give a pole discrepancy by the palaeomagnetic method. On this view the observed pole differences are therefore the outcome of either translation in a meridional direction, or of rotation.

(*c*) Assuming no relative movement between Europe and North America (for the reasons given above, relative change in longitude is not excluded) during and since the Cretaceous, the concurrence of Cretaceous and Lower Tertiary poles from these continents substantiates the dipole hypothesis for this region at this time. Assuming further that during this time there was no relative movement between these regions and Australia, it follows that the geomagnetic field, although dipolar in Europe and North America, cannot have been so in the antipodes in Australia. By the Pleistocene this disorder was lost and the average field became that of a geocentric dipole at all these places. From symmetry considerations this argument seems unlikely, and as there is already considerable theoretical (**24, 25**) and observational (**15, 22**) evidence in support of the dipole hypothesis, this point of view is regarded as tenuous although it cannot at the moment be excluded.

5.4. *Secular variation.*—In most cases in volcanic rocks, variation in directions between specimens from the same site (ω) will be due to experimental error and inaccuracies in the initial alignment. Variation between the true mean site directions (β) will in general be due to geological tilting and to secular change of the geomagnetic field during the time covered by the samples, in this case 10^6 years for the Newer Volcanics and a comparable period for the Older Volcanics. Tilting is negligible in the former, and probably unimportant in the latter, where disturbed beds have been avoided, so β provides an estimate of the range of secular variation. Watson and Irving (**10**) have given the method of separating ω and β from the data and β for both groups is given in Table I. The values for reversed and normal sites in the Newer Volcanics are similar and the agreement between the Newer and Older Volcanics, when treated as a whole, is extremely good. This agreement between analyses of many data from two geologically distinct groups gives reason for supposing that ± 23–25° is a realistic estimate of the range of secular variation over periods of time of about 10^6 years in Victoria at the times represented.

We wish to offer our grateful thanks to Dr O. P. Singleton of the Department of Geology in the University of Melbourne for his very great help with the sampling problems and other geological aspects of this work.

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1957 *May* 15.

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Notes added in proof

(1) A paper discussing the geological aspects of this work and containing all the primary data is to appear shortly in the *Proceedings of the Royal Society of Victoria.*

(2) Recent work by Cox (**26**) on lavas of Eocene age in the Western United States suggests that comparatively rapid polar wandering of the type mentioned in Section 5·3 (*a*) may have occurred in Lower Tertiary times.

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